

OPEN ENDED LAB

Electronic Light Runner

Aim

To design and implement an Electronic Light Runner circuit using 555 Timer ICs where LEDs glow sequentially with time delay created by RC components.

Problem Statement

Design a simple LED sequencing system using 555 Timer ICs, resistors, capacitors, and LEDs. The circuit should create automatic timed switching so that LEDs turn ON one after another continuously.

Components Required

- 555 Timer IC x 1Nos
- IC 7486(XOR Gate)
- IC 7408(AND Gate)
- IC 7404(NOT Gate)
- 100uF Capacitor x 1Nos
- 100k Resistor
- 47k Resistor
- 390 Ohms Resistor x 3Nos
- Breadboard
- Connecting Wires

Main Components Description

555 Timer IC: The 555 Timer IC is the main control unit of the circuit. It generates timing pulses that control the sequential blinking of LEDs.

IC 7486(XOR Gate):

Produces HIGH output when inputs are different. Used for abnormal/alternate logic.

IC 7408(AND Gate):

Produces HIGH output only when all inputs are HIGH. Used for safe condition indication.

IC 7404(NOT Gate):

Inverts the input signal. Converts HIGH to LOW and LOW to HIGH.

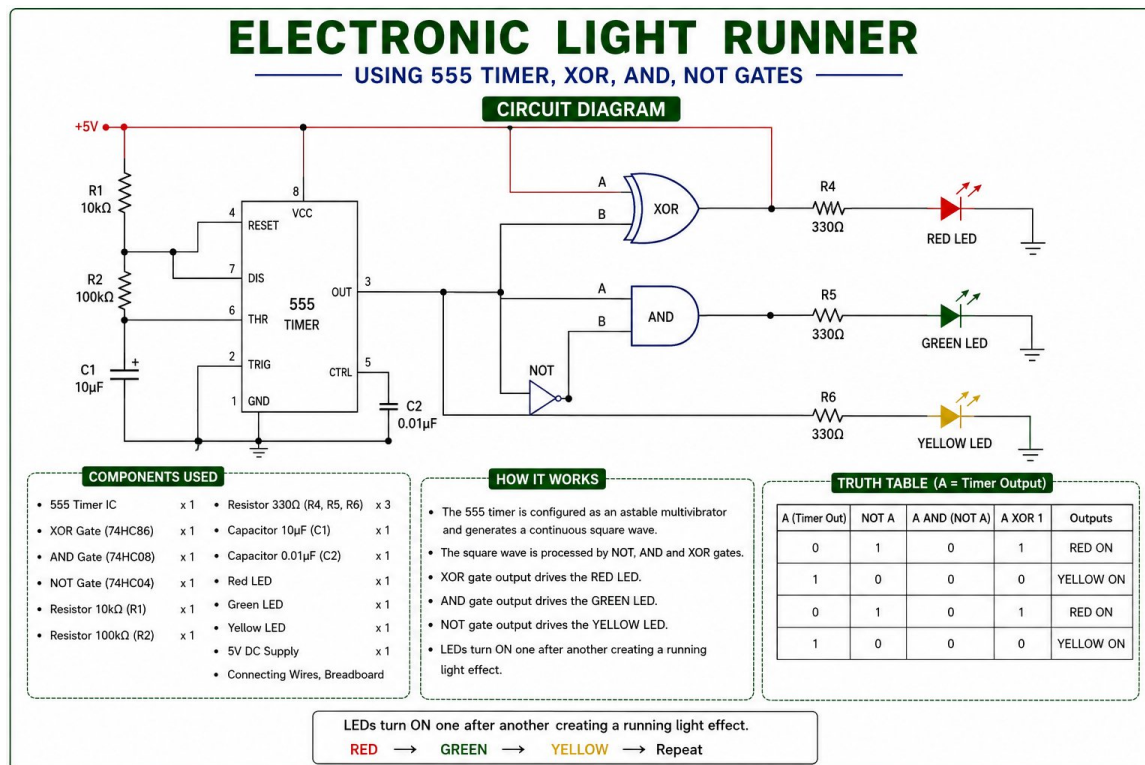
Capacitor: Capacitors store and release electrical energy to create delay in LED switching.

Resistor: Resistors control current flow and determine the charging and discharging time of capacitors.

LED: LEDs are used as visual indicators which glow sequentially.

Breadboard: The breadboard is used for temporary circuit implementation and testing.

Circuit Design



Working Principle

The circuit works based on the charging and discharging process of capacitors through resistors connected with 555 Timer ICs. Due to the RC timing delay, LEDs glow one after another creating a running light effect. The sequence repeats continuously while power is supplied.

The circuit works using a **555 Timer IC** configured in **astable mode**. The timer continuously generates square wave pulses.

1. The output pulse from the **555 Timer** is sent to the logic gates.
2. The **NOT gate** inverts the timer signal.
3. The **AND gate** checks logic conditions and drives the **Green LED**.
4. The **XOR gate** compares signals and drives the **Red LED**.
5. Another output line activates the **Yellow LED**.
6. LEDs turn ON and OFF sequentially, creating an **electronic running light effect**.

The timing speed depends on the values of the resistors and capacitors connected with the 555 Timer IC.

Applications

- Signal indication systems
- Advertising displays
- Emergency warning indicators
- Learning and demonstration of timing circuits
- Security systems
- Emergency warning systems
- Traffic signal systems
- Industrial indication systems
- Automation and control systems
- LED display boards

Conclusion

The Electronic Light Runner project successfully demonstrates sequential LED operation using 555 Timer ICs and RC delay mechanisms. The project is simple, cost-effective, and useful for understanding timing and automation concepts in electronics.